*NUST INSTITUTE OF CIVIL ENGINEERING (NICE), SECTOR H-12, ISLAMABAD*

# CE-411 Steel Structures,

# COMPLEX ENGINEERING PROBLEM

## Factory for production and storage of polythene

## [This problem will assess CLO-4: Evaluate and design various steel members /structure due to various load effects and develop a solution to address practical problems]

**Client's Requirements**

A multi-national polythene manufacturer requires a large, 200 ft. (60m) square, single-storey factory, on a green field site. The client's Architect has stipulated that one central column only is to be allowed in the factory, which is clad with profiled steel sheet sandwich panels (50% roof lighting, no windows in the sides of the factory). Apart from small personnel doors, the figure also shows the requirement for one large sliding door opening, 16 ft. (5m) high, and a minimum clearance inside the building of 20ft. (6m) Only the major structural members are offered as calculation in this example, and only selected important connections are designed. Also, several checks normally included in a full design, such as secondary bending and local buckling of the truss member walls, have been omitted.

Roof loading (offices and factory) – 0.75 kN/m2

Factory roof services - 0.65 kN/m2

Loading – wind load

Basic wind speed 46m/sec (outskirts of a town- level site)

Soil safe bearing capacity

Stiff clay, safe bearing 250 kN/m2

Design Aids

The roof purlins can be taken from manufacturers design tables

Prepare a design appraisal with appropriate sketches indicating a viable solution for the proposed structure including the foundations. Indicate clearly the functional framing, load transfer, serviceability, and stability aspects. Review and critically appraise the proposed scheme. **(30 Marks)**

- A technical note narrating the client requirement and how these will be addressed (5 marks)
- Concept Sketches showing functional framing, load transfer, stability of the structure (20 marks)
- A statement about the proposed scheme and its advantages and disadvantages (5 marks)

For the proposed scheme, prepare sufficient design calculations to establish the form and size of all the structural elements including the foundations. **(35 Marks)**

- Prepare design calculation for critical elements; i.e. the main load bearing slabs, beams, columns and foundation loads. (35 marks)

Prepare general arrangement drawings, which may include plans, section and elevations, to show the dimensions, layout and disposition of the structural elements. **(35 Marks)**

- Prepare layout drawings, plans, sections and elevations (35 marks)